

Linux LVM Storage Extension Guide

Step-by-step technical documentation for expanding root logical volumes and XFS filesystems.

Target Environment: Oracle Linux 7 / RHEL 7 | **Objective:** Add new disk (/dev/sdb), create partition, extend volume group (ol), expand root logical volume, and grow XFS filesystem live.

1. Initial Disk Layout Analysis

Inspect existing block devices and logical volume layouts using `fdisk` and `lsblk` to verify available space and target targets.

Command: Check Partition Tables & Disk Details

```
[root@localhost ~]# fdisk -l

Disk /dev/sda: 75.2 GB, 75161927680 bytes, 146800640 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
/dev/sda1 *          2048      2099199      1048576    83  Linux
/dev/sda2            2099200    146800639     72350720    8e  Linux LVM

Disk /dev/sdb: 32.2 GB, 32212254720 bytes, 62914560 sectors
```

2. Partitioning the New Disk

Create a new primary partition spanning the entire capacity of the newly attached 32 GB disk (/dev/sdb).

Command: Interactive fdisk Session to Create /dev/sdb1

```
[root@localhost ~]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).

Command (m for help): n
Partition type:
   p   primary (0 primary, 0 extended, 4 free)
   e   extended
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-62914559, default 2048):
Last sector, +sectors or +size{K,M,G} (2048-62914559, default 62914559):
Partition 1 of type Linux and of size 30 GiB is set

Command (m for help): w
The partition table has been altered!
```

3. Configuring LVM Physical Volume & Volume Group

Initialize the new partition as an LVM Physical Volume (PV) and extend the existing Volume Group (o1).

Command: Create PV and Extend Volume Group

```
[root@localhost ~]# pvcreate /dev/sdb1
Physical volume "/dev/sdb1" successfully created.

[root@localhost ~]# vgextend o1 /dev/sdb1
Volume group "o1" successfully extended

[root@localhost ~]# vgsdisplay o1 | grep Free
Free  PE / Size      7680 / 30.00 GiB
```

4. Expanding the Logical Volume

Extend the root logical volume (/dev/o1/root) utilizing 100% of the newly added free space in the volume group.

Command: Resize Logical Volume

```
[root@localhost ~]# lvextend -l +100%FREE /dev/o1/root
Size of logical volume o1/root changed from 43.08 GiB (11029 extents) to 73.08 GiB
(18709 extents).
Logical volume o1/root successfully resized.
```

5. Updating the XFS Filesystem

Upgrade xfsprogs to the compatible version and resize the live XFS filesystem to incorporate the expanded logical volume capacity.

Command: Update xfsprogs and Grow Filesystem

```
[root@localhost ~]# yum install xfsprogs.x86_64 --assumeyes
...
Updating      :
xfsprogs-5.4.0-1.0.1.el7.x86_64
                                         1/2

Cleanup       :
xfsprogs-4.5.0-22.0.1.el7.x86_64
                                         2/2

[root@localhost ~]# xfs_growfs /dev/o1/root
data blocks changed from 11293696 to 19158016
```

6. Verification

Confirm that the root mount point now correctly reflects the expanded storage capacity.

Command: Check Mounted Filesystems (df -h)

```
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        1.8G   0    1.8G   0% /dev
tmpfs           1.8G   0    1.8G   0% /dev/shm
/dev/mapper/ol-root 74G   28G   46G   38% /
/dev/sda1       1014M 235M  780M  24% /boot
/dev/mapper/ol-home 22G   37M   21G   1% /home
```

Summary of Operation

Component	Original Size	New Size	Action Taken
New Disk (/dev/sdb)	32.2 GB	32.2 GB	Created primary partition /dev/sdb1 (30 GiB)
Volume Group (ol)	~69.00 GiB	~99.00 GiB	Extended via vgextend ol /dev/sdb1
Root Logical Volume	43.08 GiB	73.08 GiB	Extended via lvextend -l +100%FREE
XFS Filesystem	43.08 GiB	74.00 GiB	Resized online via xfs_growfs